S.No: 1	Exp. Name: ***Registration Form using Table***	**Date: 2026-08-12**

Aim:
Build a registration form laid out inside an HTML table.

Form Fields

- **Name**: A text input with `name="name"`.
- **Password**: A password input with `name="password"`.
- **E-mail**: An email input with `name="email"`.
- **Phone Number**: A telephone input with `type="tel"` and `name="phone"`.
- **Sex**: Two radio buttons for "Male" and "Female", both with `name="sex"`.
- **Date of Birth**: Three dropdowns with `name="day"`, `name="month"`, and `name="year"`.
- **Languages Known**: Four checkboxes for English, Telugu, Hindi, and Tamil, each with a matching lowercase `name` attribute.
- **Address**: A textarea with `name="address"`.

Table Layout

- Place each label in one column and its input in the adjacent column.
- Set the table border to `1`.

Buttons

- A <button> element with `type="submit"` and the text "Submit".
- A <button> element with `type="reset"` and the text "Reset".

The form should look like this.

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Registration Form

Name:	
Password:	
E-mail:	
Phone Number:	
Sex:	○ Male ○ Female
Date of Birth:	Day ▾ Month ▾ Year ▾
Languages Known:	☐ English ☐ Telugu ☐ Hindi ☐ Tamil
Address:	
Submit Reset	

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Registration Form</title>
</head>
<body>
<h2>Registration Form</h2>
<!--Write your code here...-->
<form>
        <table border="1">
                <tr>
                        <td>Name:  </td>
                        <td><input type="text" name="name"></td>
                </tr>
                <tr>
                        <td>Password:</td>
                        <td><input type="password" name="password">
</td>
                </tr>
                <tr>
                        <td>E-mail:</td>
                        <td><input type="email" name="email"></td>
                </tr>
                <tr>
                        <td>Phone Number:</td>
                        <td><input type="tel" name="phone"></td>
                </tr>
                <tr>
                        <td>Sex:</td>
                        <td><input type="radio" name="sex"
value="male">Male<br>
                        <input type="radio" name="sex"
value="female">Female<br></td>
                </tr>
                <tr>
                        <td>Date of Birth:</td>
                        <td>
                                <select name="day">
                                        <option>Day</option>
                                        <option>1</option>
<option>2</option><option>3</option>
```

```
                                </select>
                                <select name="month">
                                        <option>Month</option>
                                        <option>January</option>
<option>February</option><option>March</option><option>April</option>
</select>
                                <select name="year">
                                        <option>Year</option>
                                        <option>2005</option>
<option>2006</option><option>2007</option><option>2008</option>
</select>
                        </td>
                </tr>
                <tr>
                        <td>Languages Known:</td>
                        <td><input type="checkbox" name="english"
value="english">English<br>
                                <input type="checkbox" name="telugu"
value="telugu">Telugu<br>
                                <input type="checkbox" name="hindi"
value="hindi">Hindi<br>
                                <input type="checkbox" name="tamil"
value="tamil">Tamil</td>
                </tr>
                <tr>
                        <td>Address:</td>
                        <td><textarea name="address" rows="4" cols="25"
>
                        </textarea>
                        </td>
                </tr>
                <tr>
                        <td colspan="2">
                                <button type="submit">Submit</button>
                                <button type="reset">Reset</button>
</td>
                </tr>




</body>
</html>
```

S.No: 2	Exp. Name: ***Ravi's Portfolio Page***	**Date: 2026-08-28**

Aim:

Ravi is designing his personal portfolio webpage. He wants each section to look visually distinct. Style the text in each section as described below.

- In the `<header>`, the `<h1>` should have `font-family: Verdana`, `color: royalblue`, `text-shadow: 3px 3px 8px gray`, and `text-align: center`.
- In the `<main>`, the `<h2>` should have `text-transform: uppercase`, `color: darkred`, and `letter-spacing: 2px`.
- In the `<main>`, the `<p>` should have `font-size: 16px`, `color: #333`, and `line-height: 1.7`.
- In the `<footer>`, the `<p>` should have `text-align: center`, `font-style: italic`, and `color: darkgreen`.

Source Code:

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index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-
scale=1.0"/>
  <title>Ravi’s Portfolio Page</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header>
    <h1>Welcome to My Portfolio</h1>
  </header>

  <main>
    <h2>about me</h2>
    <p>Hello! I'm Ravi, a passionate front-end developer who loves
designing simple and elegant web experiences for users.</p>
  </main>

  <footer>
    <p>© 2025 Ravi Sharma | Designed with creativity and code.</p>
  </footer>
</body>
</html>
```

styles.css

```
/*Write your code here*/
header h1{
        font-family:verdana;
        color:royalblue;
        text-shadow:3px 3px 8px gray;
        text-align:center;
}
main h2{
        text-transform:uppercase;
        color:darkred;
        letter-spacing:2px;
}
main p{
        font-size:16px;
        color:#333;
        line-height:1.7;
}
footer p{
        text-align:center;
        font-style:italic;
        color:darkgreen;
}
```

S.No: 3	Exp. Name: ***Simple Calculator***	**Date: 2026-09-22**

Aim:

You are provided with an HTML file that defines the structure of a basic calculator. The calculator supports the following arithmetic operations:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Division (/)

Your task is to complete the JavaScript file (script.js) so that the calculator functions correctly.

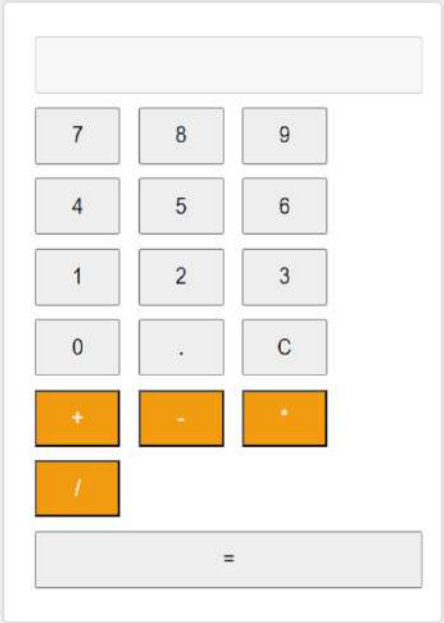
Provided Files

HTML File (index.html)

The HTML file already includes:

- A display area to show the current input or result.
- Buttons for digits (0–9).
- A decimal point button (.).
- Operation buttons (+, −, *, /).
- An equals button (=).
- A clear button (C).

You do not need to modify the HTML file.



Complete the JavaScript Functionality:

You must complete the logic in **script.js** by implementing the following functions correctly:

1. appendNumber(number)

This function is called when a number button is clicked.
It should:

- Allow multi-digit number input.
- Append the clicked digit to the current input.
- Handle decimal input properly (prevent multiple decimals in a single number).

2. setOperation(operation)

This function is called when an arithmetic operator (+, −, *, /) is clicked.
It should:

- Store the current input as the previous operand.
- Store the selected operation.
- Prepare the calculator to receive the next number.
- Prevent setting an operation if there is no current input.

3. calculate()

This function is called when the equals (=) button is clicked.
It should:

- Perform the selected arithmetic operation using the stored previous value and current value.
- Display the result.
- Handle division by zero by displaying:

```
Cannot divide by zero
```

If either the previous value or current value is empty, display:

```
Error
```

4. clearDisplay()

This function is called when the clear (C) button is clicked.
It should:

- Reset the current input.
- Reset the previous input.
- Reset the selected operation.
- Clear the display.

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Simple Calculator</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            background-color: #f4f4f4;
        }
        .calculator {
            border: 1px solid #ccc;
            padding: 20px;
            border-radius: 5px;
            background-color: #fff;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
        }
        .calculator input, .calculator button {
            margin: 5px;
            padding: 10px;
            font-size: 16px;
        }
        .calculator input {
            width: 100%;
            box-sizing: border-box;
        }
        .calculator button {
            width: 22%;
        }
        .calculator button.operator {
            background-color: #f39c12;
            color: #fff;
        }
    </style>
</head>
<body>
    <div class="calculator">
        <input type="text" id="display" disabled>
```

```
        <br>
        <button onclick="appendNumber('7')">7</button>
        <button onclick="appendNumber('8')">8</button>
        <button onclick="appendNumber('9')">9</button>
        <br>
        <button onclick="appendNumber('4')">4</button>
        <button onclick="appendNumber('5')">5</button>
        <button onclick="appendNumber('6')">6</button>
        <br>
        <button onclick="appendNumber('1')">1</button>
        <button onclick="appendNumber('2')">2</button>
        <button onclick="appendNumber('3')">3</button>
        <br>
        <button onclick="appendNumber('0')">0</button>
        <button onclick="appendNumber('.')">.</button>
        <button onclick="clearDisplay()">C</button>
        <button class="operator" onclick="setOperation('+')">+</button>
        <button class="operator" onclick="setOperation('-')">-</button>
        <button class="operator" onclick="setOperation('*')">*</button>
        <button class="operator" onclick="setOperation('/')">/</button>
        <br>
        <button style="width: 100%;" onclick="calculate()">=</button>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

script.js

```
let currentInput = '';
let previousInput = '';
let operation = null;

function appendNumber(number) {
        if (number === '.'&& currentInput.includes('.')){
                return;
        }
        if(number === '.' && currentInput === ''){
                currentInput = '0.';
        }else{
                currentInput += number;
        }
        updateDisplay();

}

function setOperation(op) {
        if (currentInput === ''){
                return;
        }
        previousInput = currentInput;
        operation = op;
        currentInput ='';

        updateDisplay();
}

function clearDisplay() {
        currentInput = '';
        previousInput = '';
        operation = null;

        updateDisplay();
}

function calculate() {
        if(previousInput === ''|| currentInput === ''){
                printToDisplay('Error');
                return;

        }
        let prev = Number(previousInput);
        let current = Number(currentInput);
```

```
        let result;

        if(operation==='+'){
                result=prev+current;
        }else if(operation==='-'){
                result=prev-current;
        }else if(operation==='*'){
                result=prev*current;
        }else if(operation==='/'){
                if(current===0){
                        printToDisplay('Cannot divide by zero');
                        return;
                }
                result=prev/current;
        }else{
                printToDisplay('Error');
                return;
        }

        currentInput = String(result);
        previousInput = '';
        operation = null;

        updateDisplay();
}

function updateDisplay() {
    document.getElementById('display').value = currentInput;
}

function printToDisplay(message) {
    document.getElementById('display').value = message;
}
```

S.No: 4	Exp. Name: ***Dynamic To-Do List with Modals***	**Date: 2026-09-13**

Aim:

You have joined a front-end team building a productivity app. The HTML and CSS for a To-Do List app are already provided. Write the JavaScript in `script.js` to make the app interactive.

Your Task

1. Add Task

When the user types a task and clicks the `Add` button, add a new `<li>` to the list with `id="task-list"`. Each item must include:

- A `<span>` holding the task text
- Two buttons labelled `Edit` and `Delete`

2. Edit Task with Modal

Clicking `Edit` on a task should:

- Open `#edit-modal` by setting its `display` to `block`
- Show the current task text inside `#edit-input`
- Update the task text when the user clicks `Save`

The modal should close when:

- The user clicks `Save`
- The user clicks the close icon at the top right
- The user clicks outside the modal

3. Delete Task with Confirmation

Clicking `Delete` on a task should:

- Open `#delete-modal` by setting its `display` to `block`
- Remove the task if the user clicks `Yes`
- Close the modal without deleting if the user clicks `No`
- Close the modal if the user clicks outside the confirmation box

Note:

- Trigger the **Add** action when the user clicks the **Add** button. You may implement this using either the button's *click* event or the form's submit event (using `event.preventDefault()` if needed).

Source Code:

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index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-
scale=1.0"/>
  <title>Dynamic To-Do List</title>
  <link rel="stylesheet" href="style.css" />
</head>
<body>
  <div class="container">
    <h1>To-Do List</h1>
    <form id="task-form">
      <input type="text" id="task-input" placeholder="Add a new task"
required />
      <button type="submit" id="add-task">Add</button>
    </form>
    <ul id="task-list"></ul>
  </div>

  <!-- Edit Modal -->
  <div id="edit-modal" class="modal">
    <div class="modal-content">
      <span class="close-btn" id="close-edit-modal">&times;</span>
      <h2>Edit Task</h2>
      <input type="text" id="edit-input" />
      <button id="save-edit">Save</button>
    </div>
  </div>

  <!-- Delete Confirmation Modal -->
  <div id="delete-modal" class="modal">
    <div class="modal-content">
      <p>Are you sure you want to delete this task?</p>
      <button id="confirm-delete">Yes</button>
      <button id="cancel-delete">No</button>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

script.js

```
const taskForm = document.getElementById("task-form");
const taskInput = document.getElementById("task-input");
const taskList = document.getElementById("task-list");

const editModal=document.getElementById("edit-modal");
const editInput=document.getElementById("edit-input");
const saveEdit = document.getElementById("save-edit");
const closeEditModal = document.getElementById("close-edit-modal");

const deleteModal = document.getElementById("delete-modal");
const confirmDelete=document.getElementById("confirm-delete");
const cancelDelete=document.getElementById("cancel-delete");


let currentTask=null;


taskForm.addEventListener("submit",function(event){
        event.preventDefault();

        const taskText=taskInput.value.trim();

        if(taskText===""){
                return;
        }

        const li=document.createElement("li");

        const span=document.createElement("span");
        span.textContent=taskText;

        const editButton = document.createElement("button");
        editButton.textContent = "Edit";
        editButton.type="button";

        const deleteButton=document.createElement("button");
        deleteButton.textContent="Delete";
        deleteButton.type="button";

        li.appendChild(span);
        li.appendChild(editButton);
        li.appendChild(deleteButton);

        taskList.appendChild(li);
```

```
        taskInput.value="";
});

taskList.addEventListener("click",function(event){
        if(!event.target.matches("button")){
                return;
        }
        const button = event.target;
        if(button.textContent === "Edit"){
                currentTask = button.parentElement;
                const span = currentTask.querySelector("span");
                editInput.value = span.textContent;
                editModal.style.display = "block";
        }
        if(button.textContent==="Delete"){
                currentTask=button.parentElement;
                deleteModal.style.display="block";
        }
});

saveEdit.addEventListener("click",function(){
        if(currentTask){
                const newText=editInput.value.trim();

                if(newText !==""){

currentTask.querySelector("span").textContent=newText;
                }
                editModal.style.display="none";
                currentTask=null;
        }
});
closeEditModal.addEventListener("click",function(){
        editModal.style.display="none";
        currentTask=null;
});
confirmDelete.addEventListener("click",function(){
        if(currentTask){
                currentTask.remove();
        }
        deleteModal.style.display="none";
        currentTask = null;
});
cancelDelete.addEventListener("click",function(){
        deleteModal.style.display="none";
```

```
        currentTask=null;
});
window.addEventListener("click",function(event){
        if(event.target===editModal){
                editModal.style.display="none";
                currentTask=null
        }
        if(event.target===deleteModal){
                deleteModal.style.display="none";
                currentTask=null;
        }
});
```

S.No: 5	Exp. Name: ***Node.js Event Logger Using Packages, Events, Timers, and File I/O***	**Date: 2026-09-29**

Aim:

In this experiment, you will build a Node.js event logging system that connects four core concepts into one working flow. The `server.js`, browser UI, and CSS file are already provided in the default code. Your only task is to implement two things inside `logger.js`.

Your Task

Implement the following inside `logger.js`:

- Create a class **LogEmitter** that extends the built-in `EventEmitter` class.
- Implement the **startLogging** function. It must do the following:
    - Create an instance of **LogEmitter**.
- Register a listener on the `"log"` event. Each time the event fires, the listener must build a log entry in this exact format and append it to `LOG_FILE` using `fs.appendFileSync`:
- `[<uuid>] <timestamp> — <message>\n`
- where `uuid` comes from `uuidv4()` and `timestamp` comes from `new Date().toISOString()`.Use `setInterval` to emit the `"log"` event every `2000` ms with the message `"Event fired"`.Use `clearInterval` to stop the interval after `5` ticks.Emit a `"done"` event on the emitter after the 5th tick.Return the **LogEmitter** instance.

Note: Click **Start** in the browser UI to trigger your implementation and observe the live dashboard. Click **Reset** to clear and run again.

Source Code:

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my-app/logger.js

```
const EventEmitter = require("events");
const fs = require("fs");
const path = require("path");
const { v4: uuidv4 } = require("uuid");

const LOG_FILE = path.join(__dirname, "logs", "events.log");

if (!fs.existsSync(path.join(__dirname, "logs"))) {
  fs.mkdirSync(path.join(__dirname, "logs"));
}

// ── TASK 1: Create a custom EventEmitter
──────────────────────────────────

// Create a class LogEmitter that extends EventEmitter.
// Write your code here
class LogEmitter extends EventEmitter{

}

// ── TASK 2: Implement the startLogging function
──────────────────────────

// This function must:
//   1. Create an instance of LogEmitter
//   2. Register a listener on the "log" event that:
//        a. Builds a log entry in this exact format:
//           [<uuid>] <timestamp> — <message>\n
//           where uuid comes from uuidv4() and timestamp from new
Date().toISOString()
//        b. Appends the entry to LOG_FILE using fs.appendFileSync
//   3. Use setInterval to emit the "log" event every 2000ms with the
message "Event fired"
//   4. Use clearInterval to stop the interval after 5 ticks
//   5. Emit a "done" event on the emitter after the 5th tick
//   6. Return the LogEmitter instance
function startLogging() {
        const myLogger = new LogEmitter();
        myLogger.on('log',(data)=>{
                fs.appendFileSync(LOG_FILE,`[${uuidv4()}] ${new
Date().toISOString()} \u2014 ${data}\n`);
        });
        let count = 0;
        const intervalId = setInterval(()=>{
                myLogger.emit('log','Event fired');
                count++;
```

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```
                if(count === 5){
                        clearInterval(intervalId);
                        myLogger.emit('done');
                }
        }, 2000);
        return emitter;
  // Write your code here
}

module.exports = { startLogging, LOG_FILE };
```

my-app/package.json

```
{
  "name": "my-app",
  "version": "1.0.0",
  "description": "Node.js Events, Timers, Packages and File I/O",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "uuid": "^9.0.0"
  }
}
```

my-app/public/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-
scale=1.0"/>
  <title>Node.js Event Logger</title>
  <link rel="stylesheet" href="/style.css" />
</head>
<body>

  <div class="page-header">
    <h1>Node.js Event Logger</h1>
    <p class="subtitle">Monitor live event activity powered by your
Node.js implementation</p>
  </div>

  <div class="main">

    <div class="stat-grid">
      <div class="stat-card">
        <div class="stat-label">Status</div>
        <div class="stat-value" id="stat-status">
          <span class="badge idle">Idle</span>
        </div>
      </div>
      <div class="stat-card">
        <div class="stat-label">Ticks Fired</div>
        <div class="stat-value big" id="stat-ticks">0 <span
class="of">/ 5</span></div>
      </div>
      <div class="stat-card">
        <div class="stat-label">Last Event Time</div>
        <div class="stat-value mono" id="stat-time">—</div>
      </div>
      <div class="stat-card">
        <div class="stat-label">Log File Size</div>
        <div class="stat-value mono" id="stat-size">0 bytes</div>
      </div>
    </div>

    <div class="card">
      <div class="card-title">Last Generated UUID</div>
      <div class="uuid-display" id="stat-uuid">—</div>
```

```
    </div>

    <div class="card">
      <div class="card-title">Event Log</div>
      <div class="empty-state" id="empty-state">
        <span class="empty-icon">◯</span>
        <p>No events yet. Click Start to begin.</p>
      </div>
      <div class="log-feed" id="log-feed"></div>
    </div>

    <div class="btn-row">
      <button id="start-btn" onclick="handleStart()">Start</button>
      <button id="reset-btn" class="btn-secondary"
onclick="handleReset()">Reset</button>
    </div>

  </div>

  <script>
    let pollInterval = null;
    let lastCount = 0;

    function setStatus(type, label) {
      document.getElementById("stat-status").innerHTML = '<span
class="badge ' + type + '">' + label + '</span>';
    }

    window.handleStart = async function () {
      document.getElementById("start-btn").disabled = true;
      document.getElementById("reset-btn").disabled = true;
      setStatus("running", "Running");
      try {
        const res = await fetch("/start", { method: "POST" });
        const data = await res.json();
        if (data.status === "error") {
          setStatus("error", "Error");
          document.getElementById("stat-uuid").textContent =
data.message;
          document.getElementById("start-btn").disabled = false;
          document.getElementById("reset-btn").disabled = false;
          return;
        }
        document.getElementById("empty-state").style.display = "none";
        lastCount = 0;
        pollInterval = setInterval(pollLogs, 600);
```

```
      } catch (err) {
        setStatus("error", "Error");
        document.getElementById("start-btn").disabled = false;
        document.getElementById("reset-btn").disabled = false;
      }
    };

    window.handleReset = async function () {
      clearInterval(pollInterval);
      await fetch("/reset", { method: "POST" });
      lastCount = 0;
      document.getElementById("stat-ticks").innerHTML = '0 <span
class="of">/ 5</span>';
      document.getElementById("stat-time").innerHTML = "—";
      document.getElementById("stat-size").textContent = "0 bytes";
      document.getElementById("stat-uuid").innerHTML = "—";
      document.getElementById("log-feed").innerHTML = "";
      document.getElementById("empty-state").style.display = "flex";
      document.getElementById("start-btn").disabled = false;
      document.getElementById("reset-btn").disabled = false;
      setStatus("idle", "Idle");
    };

    async function pollLogs() {
      try {
        const res = await fetch("/logs");
        const data = await res.json();
        const lines = data.lines || [];

        if (lines.length !== lastCount) {
          lastCount = lines.length;
          document.getElementById("stat-ticks").innerHTML =
lines.length + ' <span class="of">/ 5</span>';
          document.getElementById("stat-size").textContent =
data.fileSize + " bytes";

          if (lines.length > 0) {
            const lastLine = lines[lines.length - 1];
            const uuidMatch = lastLine.match(/\[([^\]]+)\]/);
            const timeMatch = lastLine.match(/\]\s+([^\s]+)\s+/);
            if (uuidMatch) document.getElementById("stat-
uuid").textContent = uuidMatch[1];
            if (timeMatch) document.getElementById("stat-
time").textContent = timeMatch[1].replace("T", " ").replace("Z", "
UTC");
          }
```

```
          const feed = document.getElementById("log-feed");
          feed.innerHTML = lines.map(function(line, i) {
            const uuidMatch = line.match(/\[([^\]]+)\]/);
            const timeMatch = line.match(/\]\s+([^\s]+)\s+/);
            const msgMatch  = line.match(/\u2014\s+(.*)/);
            const uuid = uuidMatch ? uuidMatch[1] : "—";
            const time = timeMatch ? timeMatch[1].replace("T", "
").replace("Z", " UTC") : "—";
            const msg  = msgMatch  ? msgMatch[1] : line;
            return '<div class="log-row">'
              + '<span class="log-num">#' + (i + 1) + '</span>'
              + '<div class="log-details">'
              + '<span class="log-msg">' + msg + '</span>'
              + '<span class="log-meta">' + time + '</span>'
              + '<span class="log-uuid">' + uuid + '</span>'
              + '</div></div>';
          }).join("");
          feed.scrollTop = feed.scrollHeight;
        }

        if (!data.isLogging && lines.length >= 5) {
          clearInterval(pollInterval);
          setStatus("done", "Done");
          document.getElementById("start-btn").disabled = false;
          document.getElementById("reset-btn").disabled = false;
        }

        if (!data.isLogging && lines.length === 0) {
          clearInterval(pollInterval);
          setStatus("error", "Error");
          document.getElementById("start-btn").disabled = false;
          document.getElementById("reset-btn").disabled = false;
        }

      } catch (err) {
        clearInterval(pollInterval);
      }
    }
  </script>

</body>
</html>
```

my-app/public/style.css

```
* { box-sizing: border-box; margin: 0; padding: 0; }

body {
  font-family: "Segoe UI", sans-serif;
  background: #f0f2f5;
  min-height: 100vh;
  color: #1a202c;
}

.page-header {
  background: #fff;
  border-bottom: 1px solid #e2e8f0;
  padding: 24px 40px;
  text-align: center;
}

.page-header h1 {
  font-size: 1.4rem;
  font-weight: 700;
  color: #1a202c;
  margin-bottom: 4px;
}

.subtitle { font-size: 0.83rem; color: #718096; }

.main {
  max-width: 680px;
  margin: 32px auto;
  padding: 0 24px;
  display: flex;
  flex-direction: column;
  gap: 20px;
}

.stat-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 14px;
}

.stat-card {
  background: #fff;
  border: 1px solid #e2e8f0;
  border-radius: 12px;
```

```
  padding: 18px 20px;
}

.stat-label {
  font-size: 0.72rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.07em;
  color: #a0aec0;
  margin-bottom: 10px;
}

.stat-value { font-size: 0.95rem; font-weight: 600; color: #2d3748; }

.stat-value.big {
  font-size: 1.8rem;
  font-weight: 700;
  color: #1a202c;
  line-height: 1;
}

.stat-value.mono {
  font-family: monospace;
  font-size: 0.82rem;
  color: #4a5568;
  word-break: break-all;
}

.of { font-size: 1rem; color: #a0aec0; font-weight: 500; }

.badge {
  display: inline-block;
  padding: 3px 10px;
  border-radius: 20px;
  font-size: 0.75rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.05em;
}

.badge.idle    { background: #edf2f7; color: #718096; }
.badge.running { background: #ebf8ff; color: #2b6cb0; border: 1px solid
#90cdf4; }
.badge.done    { background: #f0fff4; color: #276749; border: 1px solid
#9ae6b4; }
.badge.error   { background: #fff5f5; color: #c53030; border: 1px solid
```

```
#feb2b2; }

.card {
  background: #fff;
  border-radius: 12px;
  border: 1px solid #e2e8f0;
  padding: 20px 24px;
}

.card-title {
  font-size: 0.72rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.07em;
  color: #a0aec0;
  margin-bottom: 12px;
}

.uuid-display {
  font-family: monospace;
  font-size: 0.88rem;
  color: #3182ce;
  font-weight: 600;
  background: #ebf8ff;
  border-radius: 7px;
  padding: 10px 14px;
  word-break: break-all;
  letter-spacing: 0.03em;
}

.empty-state {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  gap: 10px;
  padding: 32px 20px;
  text-align: center;
}

.empty-icon { font-size: 2rem; color: #e2e8f0; line-height: 1; }
.empty-state p { font-size: 0.83rem; color: #a0aec0; }

.log-feed {
  display: flex;
  flex-direction: column;
```

```
  gap: 8px;
  max-height: 280px;
  overflow-y: auto;
}

.log-row {
  display: flex;
  align-items: flex-start;
  gap: 12px;
  background: #f7fafc;
  border-radius: 8px;
  padding: 10px 14px;
  border-left: 3px solid #4299e1;
}

.log-num {
  font-size: 0.72rem;
  font-weight: 700;
  color: #a0aec0;
  min-width: 24px;
  padding-top: 2px;
}

.log-details {
  display: flex;
  flex-direction: column;
  gap: 3px;
  flex: 1;
}

.log-msg { font-size: 0.88rem; font-weight: 600; color: #2d3748; }
.log-meta { font-size: 0.75rem; font-family: monospace; color: #718096;
}
.log-uuid { font-size: 0.72rem; font-family: monospace; color: #a0aec0;
word-break: break-all; }

.btn-row { display: flex; gap: 12px; }

button {
  flex: 1;
  padding: 11px;
  background: #3182ce;
  color: #fff;
  border: none;
  border-radius: 8px;
  font-size: 0.9rem;
```

```
  font-weight: 600;
  cursor: pointer;
  transition: background 0.2s, opacity 0.2s;
}

button:hover   { background: #2b6cb0; }
button:disabled { opacity: 0.5; cursor: not-allowed; }

.btn-secondary {
  background: #fff;
  color: #4a5568;
  border: 1.5px solid #e2e8f0;
}

.btn-secondary:hover { background: #f7fafc; }
```

my-app/server.js

```
const http = require("http");
const fs = require("fs");
const path = require("path");

const PORT = 3000;

const SERVER_FILE = path.join(__dirname, "logger.js");
let lastMtime = fs.statSync(SERVER_FILE).mtimeMs;
setInterval(() => {
  try {
    const mtime = fs.statSync(SERVER_FILE).mtimeMs;
    if (mtime !== lastMtime) {
      console.log("logger.js changed — restarting...");
      process.exit(1);
    }
  } catch (e) {}
}, 1000);

function serveStatic(res, filePath, contentType) {
  fs.readFile(filePath, (err, data) => {
    if (err) {
      res.writeHead(404, { "Content-Type": "text/plain" });
      res.end("404 Not Found");
      return;
    }
    res.writeHead(200, { "Content-Type": contentType });
    res.end(data);
  });
}

let isLogging = false;

function handleRequest(req, res) {
  if (req.method === "GET" && req.url === "/") {
    serveStatic(res, path.join(__dirname, "public", "index.html"),
"text/html");
    return;
  }

  if (req.method === "GET" && req.url === "/style.css") {
    serveStatic(res, path.join(__dirname, "public", "style.css"),
"text/css");
    return;
  }
```

```
  if (req.method === "POST" && req.url === "/start") {
    if (isLogging) {
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ status: "already running" }));
      return;
    }
    try {
      delete require.cache[require.resolve("./logger")];
      const { startLogging, LOG_FILE } = require("./logger");
      if (typeof startLogging !== "function") {
        res.writeHead(500, { "Content-Type": "application/json" });
        res.end(JSON.stringify({ status: "error", message:
"startLogging is not implemented yet." }));
        return;
      }
      const logDir = path.join(__dirname, "logs");
      if (!fs.existsSync(logDir)) fs.mkdirSync(logDir);
      fs.writeFileSync(LOG_FILE, "");
      isLogging = true;
      const emitter = startLogging();
      emitter.on("done", () => { isLogging = false; });
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ status: "started" }));
    } catch (err) {
      isLogging = false;
      res.writeHead(500, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ status: "error", message: err.message
}));
    }
    return;
  }

  if (req.method === "POST" && req.url === "/reset") {
    isLogging = false;
    try {
      delete require.cache[require.resolve("./logger")];
      const { LOG_FILE } = require("./logger");
      if (fs.existsSync(LOG_FILE)) fs.writeFileSync(LOG_FILE, "");
    } catch (e) {}
    res.writeHead(200, { "Content-Type": "application/json" });
    res.end(JSON.stringify({ status: "reset" }));
    return;
  }

  if (req.method === "GET" && req.url === "/logs") {
```

```
    try {
      delete require.cache[require.resolve("./logger")];
      const { LOG_FILE } = require("./logger");
      const content = fs.existsSync(LOG_FILE) ?
fs.readFileSync(LOG_FILE, "utf8") : "";
      const lines = content.trim().split("\n").filter(Boolean);
      const fileSize = fs.existsSync(LOG_FILE) ?
fs.statSync(LOG_FILE).size : 0;
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ lines, isLogging, fileSize }));
    } catch (err) {
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify({ lines: [], isLogging: false, fileSize: 0
}));
    }
    return;
  }

  res.writeHead(404, { "Content-Type": "text/plain" });
  res.end("Not Found");
}

const server = http.createServer(handleRequest);
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

S.No: 6	Exp. Name: ***Developing a Single Page Application using RESTful APIs***	**Date: 2026-10-05**

Aim:

You are developing a Movie Management System for an online streaming platform. The application allows users to add new movies, update movie ratings, delete movies, and view all available movies.

The application behaves as a Single Page Application (SPA). Instead of reloading the page after every operation, it communicates with the Express.js server using RESTful APIs through JavaScript's `fetch()` method.

The MongoDB connection, Movie model, project configuration, HTML, CSS, and the basic JavaScript functions have already been provided. Your task is to complete the missing code in both **index.js** and **script.js** so that the application works correctly.

Your Tasks

Complete the missing code in **index.js** by implementing the following RESTful API routes.

1. Create Movie

Create a `POST` route at `/add-movie` that:

- Accepts book details (*title*, *director*, *year*, and *rating*) in JSON format.
- Stores the received data as a new **Movie** document in the database.

Responds with:

```
Movie added: <movie-title>
```

2.View Movies

Create a `GET` route at `/movies` that:

- Retrieves all movies from the database using `Movie.find()`.
- Returns the retrieved documents as JSON.
- If an error occurs, respond with status code 500 and an appropriate error message.

3. Update Movie rating

Create a `PUT` route at `/update-movie` that:

- Accepts *title* and *newRating* from the request body.
- Updates only the rating of the matching movie using the $set operator.
- If the update is successful, respond with:

```
Movie '<title>' rating updated successfully
```

- If no matching movie is found, respond with:

```
No movie found with title '<title>'
```

- If an error occurs, respond with status code `500`.

4. Delete Movie

Create a `DELETE` route at `/movies/:title` that:

- Deletes the movie whose title matches the route parameter.
- If the movie is deleted successfully, respond with: {"message": "Movie deleted successfully"}
- If no movie exists respond with: {"message": "Movie not found"}

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- If an error occurs, respond with status code `500` and an appropriate error message.

Complete the missing code in **script.js** by completing `fetch()` requests for the following functions.

- **addMovie()** → Write a `fetch()` request that sends a `POST` request to `/add-movie` with the provided movie details in JSON format.
- **updateMovie()** → Write a `fetch()` request that sends a `PUT` request to `/update-movie` with the provided title and newRating in JSON format.
- **deleteMovie()** → Write a `fetch()` request that sends a `DELETE` request to `/movies/:title` by appending the movie title from the input field, encoded using `encodeURIComponent()`, to `/movies/`.
- **viewMovies()** → Write a `fetch()` request that sends a `GET` request to `/movies` and retrieves the movie list for display.

Run the application and verify that movies can be added, deleted, updated, and retrieved successfully.

Note: Do not modify the provided application setup. Write only the missing code in **index.js** and **script.js**.

Source Code:

express-app/index.js

```
const express = require('express');
const mongoose = require('mongoose');
const path = require('path');
const Movie = require('./models/movieModel');

const app = express();
app.use(express.json());
app.set('view engine', 'ejs');
app.use(express.urlencoded({ extended: true }));
app.use(express.static(path.join(__dirname, 'public')));

let dbName = '';
let dbHost = '';
let dbPort = '';

mongoose.connect('mongodb://mongo:27017/libraryDB', {
  useNewUrlParser: true,
  useUnifiedTopology: true
})
.then(() => {
  console.log('Database connected successfully!');

  const db = mongoose.connection;
  dbName = db.name;
  dbHost = db.host;
  dbPort = db.port;
})
.catch((err) => {
  console.error('Database connection failed:', err.message);
});

app.get('/ui', (req, res) => {
  res.render('index');
});

// Write POST /add-movie
app.post('/add-movie',async(req,res)=>{
        try{
                const movie = new Movie(req.body);
                await movie.save();
                res.status(200).send(`Movie added: ${req.body.title}`);
        }
        catch(err){
                res.status(500).send(err.message);
```

```
        }
});


// Write GET /movies
app.get('/movies',async(req,res)=>{
        try{
                const movies = await Movie.find();
                res.status(200).json(movies);

        }catch(err){
                res.status(500).send(err.message);
        }
});



// Write PUT /update-movie
app.put('/update-movie',async(req,res)=>{
        try{
                const title = req.body.title;
                const newRating = req.body.newRating != undefined ?
req.body.newRating : req.body.rating;
                const result = await Movie.updateOne({title:title},
{$set: {rating: Number(newRating)}} );
                if(!result.matchedCount && !result.n){
                        return res.status(404).send(`No movie found
with title '${title}' `);
                }
                res.status(200).send(`Movie '${title}' rating updated
successfully`);
        }catch(err){
                res.status(500).send(err.message);
        }
});



// Write DELETE /movies/:title
app.delete('/movies/:title',async(req,res)=>{
        try{
                const title = req.params.title;
                const result = await Movie.deleteOne({title});
                if(!result.deletedCount && !result.n){
                        return res.status(404).json({message:"Movie not
found"});
```

```
                }
                res.status(200).json({message: "Movie deleted
successfully"});
        }catch(err){
                res.status(500).send(err.message);
        }
});




// -------------------- DATABASE DETAILS --------------------

app.get('/', (req, res) => {
  res.json({
    dbName,
    dbHost,
    dbPort
  });
});


// -------------------- SERVER --------------------

app.listen(3000, () => {
  console.log('Server running on http://localhost:3000');
});
```

express-app/public/script.js

```
function hideAll() {
    document.querySelectorAll(".page").forEach(page =>
page.style.display = "none");
}

function showPage(id) {
    hideAll();
    document.getElementById(id).style.display = "block";
    document.getElementById("output").innerHTML = "";
}

function goHome() {
    hideAll();
    document.getElementById("home").style.display = "block";
    document.getElementById("output").innerHTML = "";
}


async function addMovie() {
// write the fetch for adding movie
        const response = await fetch('/add-movie',{
                        method: 'POST',
                        headers: {
                                'Content-Type':
'application/json'
                        },



        body: JSON.stringify({
            title: document.getElementById("title").value,
            director: document.getElementById("director").value,
            year: Number(document.getElementById("year").value),
            rating: Number(document.getElementById("rating").value)
        })
                });
    document.getElementById("output").innerHTML = await
response.text();
}

async function updateMovie() {

// write the fetch for update
        const response = await fetch('/update-movie',{
```

```
                method: "PUT",
                headers: {
                            'Content-Type': 'application/json'
                },


        body: JSON.stringify({
          title: document.getElementById("updateTitle").value,
          newRating: Number(document.getElementById("newRating").value)
      })
        });
    document.getElementById("output").innerHTML = await
response.text();
}

// Write your code here for Delete Movie

async function deleteMovie() {
// write the fetch for delete
const title = document.getElementById('deleteTitle').value;

const response = await fetch('/movies/' + encodeURIComponent(title), {
        method: 'DELETE'
});
          const result = await response.json();
    document.getElementById("output").innerHTML = await
response.text();
}

async function viewMovies() {
    hideAll();
    document.getElementById("viewPage").style.display = "block";
// write the fetch to view
const response = await fetch('/movies');
const movies = await response.json();
    let table = `
    <table>
        <tr>
            <th>Title</th>
            <th>Director</th>
            <th>Year</th>
            <th>Rating</th>
        </tr>
    `;

    movies.forEach(movie => {
```

```
        table += `
        <tr>
            <td>${movie.title}</td>
            <td>${movie.director}</td>
            <td>${movie.year}</td>
            <td>${movie.rating}</td>
        </tr>
        `;

    });

    table += "</table>";
    document.getElementById("movieTable").innerHTML = table;
}
```

S.No: 7	Exp. Name: ***Creating and Nesting Function and Class Components***	**Date: 2026-09-27**

Aim:

You are developing a React application for a digital library. The application should display information about a book by creating and nesting React components. In this activity, the React application setup has already been provided. Your task is to create the **Book** function component and the **App** class component and nest the **Book** component inside the **App** component.

Your Task

1. Inside **Book.jsx**, create a function component named **Book** that returns a `<div>` with the class **book-card**, containing:
    - An `<h2>` element with the text "The Great Gatsby".
    - A paragraph element with the text "Author: F. Scott Fitzgerald".

Export the **Book** component.

2. In the **App.jsx** file:
    - Create a class component named **App** that extends **Component**.
    - Import the `Book` component from `./Book`.
    - Inside the `render()` method, display an `<h3>` element with the text "Books".
    - Create a `<div>` with the class name **library**, and nest the `<Book />` component inside it.
    - Return the JSX from the `render()` method.
    - Export the **App** component as the default export.

Expected Output:

When the application is executed, it should display:

```
Books
--------------------------------
The Great Gatsby
Author: F. Scott Fitzgerald
--------------------------------
```

where the **Book** component is nested inside the **App** component.

Note:

- Both **App.jsx** and **Book.jsx** are in the same path.
- Initially, the page will show a placeholder until the required code is completed.

Source Code:

functionAndClassComponents/src/App.jsx

```
import { Component } from "react";
// Import the Book component here.
import Book from "./Book"
import ErrorBoundary from "./components/ErrorBoundary";

// Create the App class component here...

class App extends Component{
        render(){
                return(
                        <div>
                                <h3>Books</h3>
                                <div className = "library">
                                        <Book />
                                </div>
                        </div>
                );
        }
}
export default App;
```

functionAndClassComponents/src/Book.jsx

```
// Create the Book function component here...
function Book(){
        return(
                <div className = "book-card">
                        <h2>The Great Gatsby</h2>
                        <p>Author: F. Scott Fitzgerald</p>
                </div>
        );

}
export default Book;
```

S.No: 9	Exp. Name: ***MongoDB CRUD Operations Using Mongoose***	**Date: 2026-10-05**

Aim:

You are developing a Book Management System for a library. The application stores book information in a MongoDB database using Mongoose. A Book model has already been provided. Your task is to complete the Express.js application by connecting to the database and implementing the required CRUD (Create, Read, Update, and Delete) operations.

Your Task

Open the **index.js** file and complete the missing code as described below:

1. Connect to MongoDB

Connect the application to the following MongoDB database:

`mongodb://mongo:27017/libraryDB`

Using the Mongoose connection object, store:

- the database name in `dbName`
- the database host in `dbHost`
- the database port in `dbPort`

2. Create Operation

Create a POST route at `/add-book` that:

- Accepts book details (`title`, `author`, `year`, and `price`) in JSON format.
- Stores the received data as a new **Book** document in the database.

Responds with:

`Book added: <book-title>`

3. Read Operation

Create a `GET` route at `/books` that:

- Retrieves all books from the database using `Book.find()`.
- Returns the retrieved documents as JSON.

If an error occurs, respond with status code `500` and an appropriate error message.

4. Update Operation

Create a `PUT` route at `/update-book` that:

- Accepts `title` and `newPrice` from the request body.
- Updates only the price of the matching book using the `$set` operator.

If the update is successful, respond with:

`Book '<title>' price updated successfully`

If no matching book is found, respond with:

`No book found with title '<title>'`

If an error occurs, respond with status code `500`.

5. Delete Operation

Create a `DELETE` route at `/books/:title` that:

- Deletes the book whose title matches the route parameter using:
  `Book.deleteOne({ title: req.params.title })`

- If the book is deleted successfully, respond with: { "message": "Book deleted successfully" }
- If no matching book is found, respond with status code 404 and: { "message": "Book not found" }

If an error occurs, respond with status code `500` and an appropriate error message.

Run the application and verify that books can be added, retrieved, updated, and deleted successfully.

Note: The **Book** model and project configuration have already been provided. Write only the missing code without modifying the existing application setup.

Source Code:

express-app/index.js

```
const express = require('express');
const mongoose = require('mongoose');
const Book = require('./models/bookModel');

const app = express();

app.use(express.json());
app.set('view engine', 'ejs');
app.use(express.urlencoded({ extended: true }));

let dbName = '';
let dbHost = '';
let dbPort = '';

// Write your code here to connect to MongoDB
mongoose.connect('mongodb://mongo:27017/libraryDB').then(()=>{
        dbName = mongoose.connection.name;
        dbHost = mongoose.connection.host;
        dbPort = mongoose.connection.port;
})
.catch(err=>{
        console.error(err);
});

app.get('/ui', (req, res) => {
  res.render('index');
});


// Write your code here for POST route (/add-book)
app.post('/add-book', async (req,res)=>{
        try{
                const book = new Book(req.body);
                await book.save();
                res.send(`Book added: ${book.title}`);
        }
        catch(err){
                res.status(500).send('Error adding book');
        }
});


// Write your code here for GET route (/books)
app.get('/books', async (req,res)=>{
```

```
        try{
                const books = await Book.find();
                res.json(books);
        }
        catch(err){
                res.status(500).json({message:'Error retrieving
books'});
        }
});

// Write your code here for PUT route (/update-book)
app.put('/update-book', async (req,res)=>{
        try{
                const{title,newPrice} = req.body;
                const book = await Book.findOneAndUpdate(
                        {title:title},
                        {$set:{price:newPrice}},
                        {new:true}
                );
                if(!book){
                        return res.send(`No book found with title
'${title}'`);
                }
                res.send(`Book '${title}' price updated successfully`);
        }
        catch(err){
                res.status(500).send('Error updating book');
        }
});

// Write your code here for DELETE route (/books/:title)
app.delete('/books/:title', async (req,res)=>{
        try{
                const result = await Book.deleteOne({
                        title: req.params.title

                });
                if(result.deletedCount === 0){
                        return res.status(404).json({
                                message:'Book not found'
                        });
                }
                res.json({
                        message:'Book deleted successfully'
                });
```

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```
        }
        catch(err){
                res.status(500).json({
                        message:'Error deleting book'
                });
        }
});

// -------------------- DATABASE DETAILS --------------------
app.get('/', (req, res) => {
  res.json({
    dbName,
    dbHost,
    dbPort
  });
});

// -------------------- SERVER --------------------
app.listen(3000, () => {
  console.log('Server running on http://localhost:3000');
});
```